

# Project Report

## Wildlife Information System (WIS)

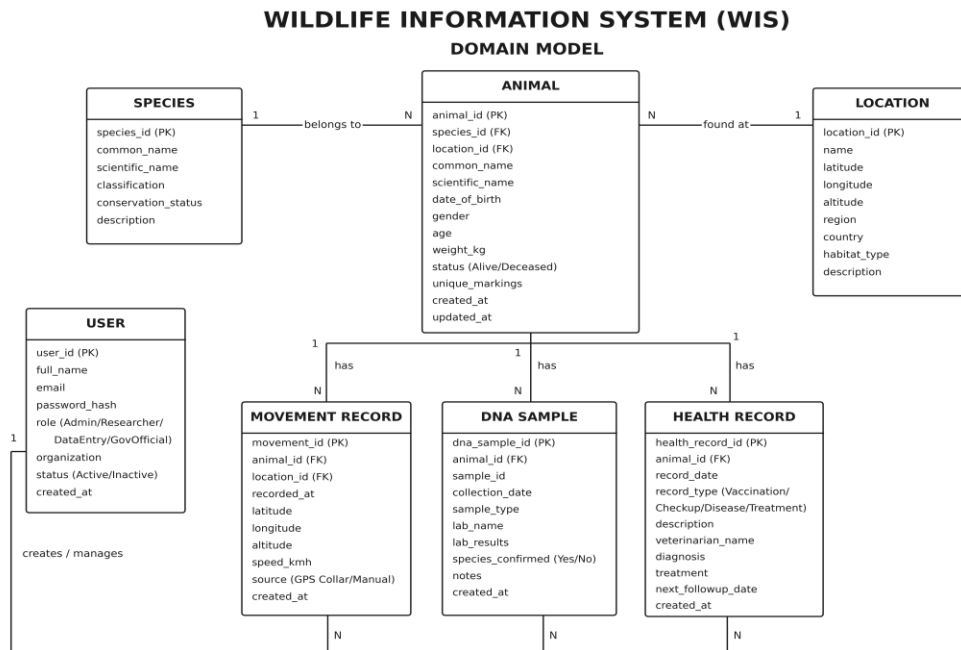
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### 3. ANALYSIS

#### 3.1 Domain Model

The domain model tells us about the core and master real world concepts each entities in the Wildlife Information System(WIS) and how they interact to each other.



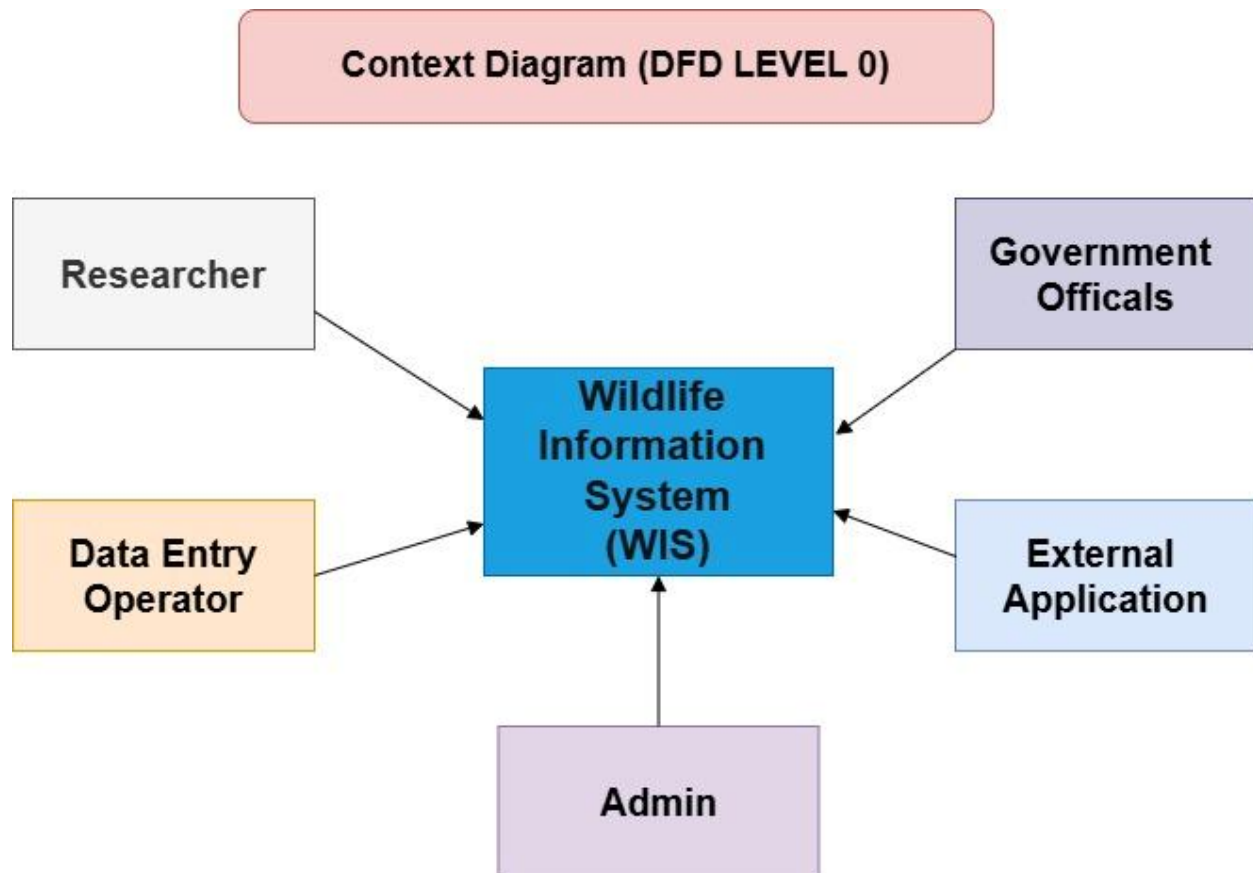
**Figure 2: Domain Model — Key Entities and Relationships**

Key entities: Animal (central), species, movement record, DNA sample, Health record, User, and Location. One Animal has many movement records, health records, and DNA samples. Each Animal belongs to one unique Species.

#### 3.2 Data Flow Diagram (DFD)

##### Level 0 DFD (Context Diagram)

The Context Diagram describes the system as a single process and identifies all external entities that interact with it.

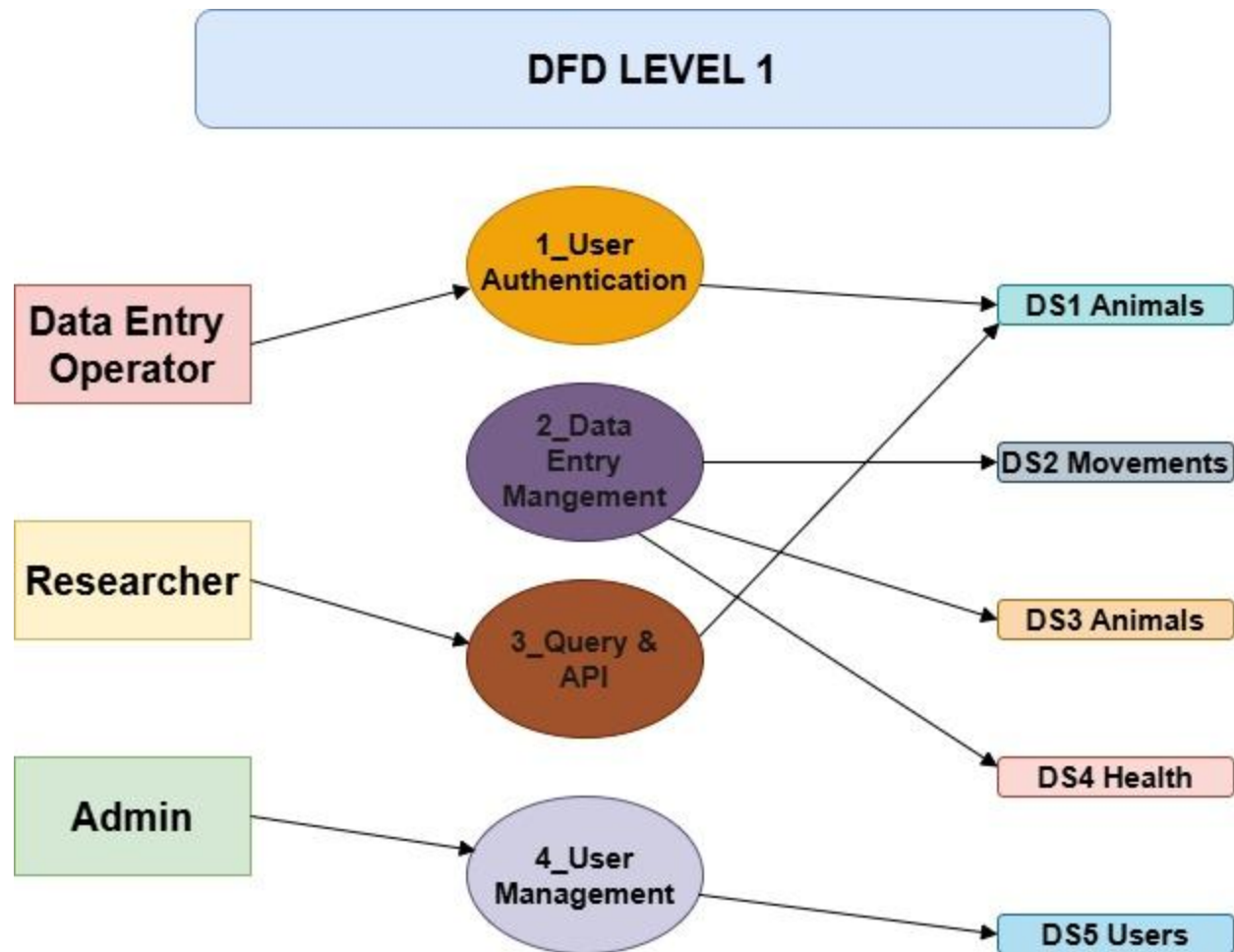


**Figure 3: Context Diagram (DFD Level 0)**

External Entities: Researcher, Data entry operator, System administrator, Government official, External application (through API). The system obtains data about animals, movement details, DNA samples, and animal health information from the Data Entry Operator. Information is requested by the Researcher and Government Officials.

### Level 1 DFD

The Level 1 DFD breaks the system into main processes:



**Figure 4: DFD Level 1 — System Processes**

**Process 1 - User Authentication:** responsible for log-in, token creation, and user-role authorization/authentication.

**Process 2 - Data Entry Processing:** accepts and processes inputted data on wildlife, including animal details, movement, DNA, and health data etc.

**Process 3 - Data Search & API Request Processing:** Accepts requests from researchers or other applications and delivers requested data from the database.

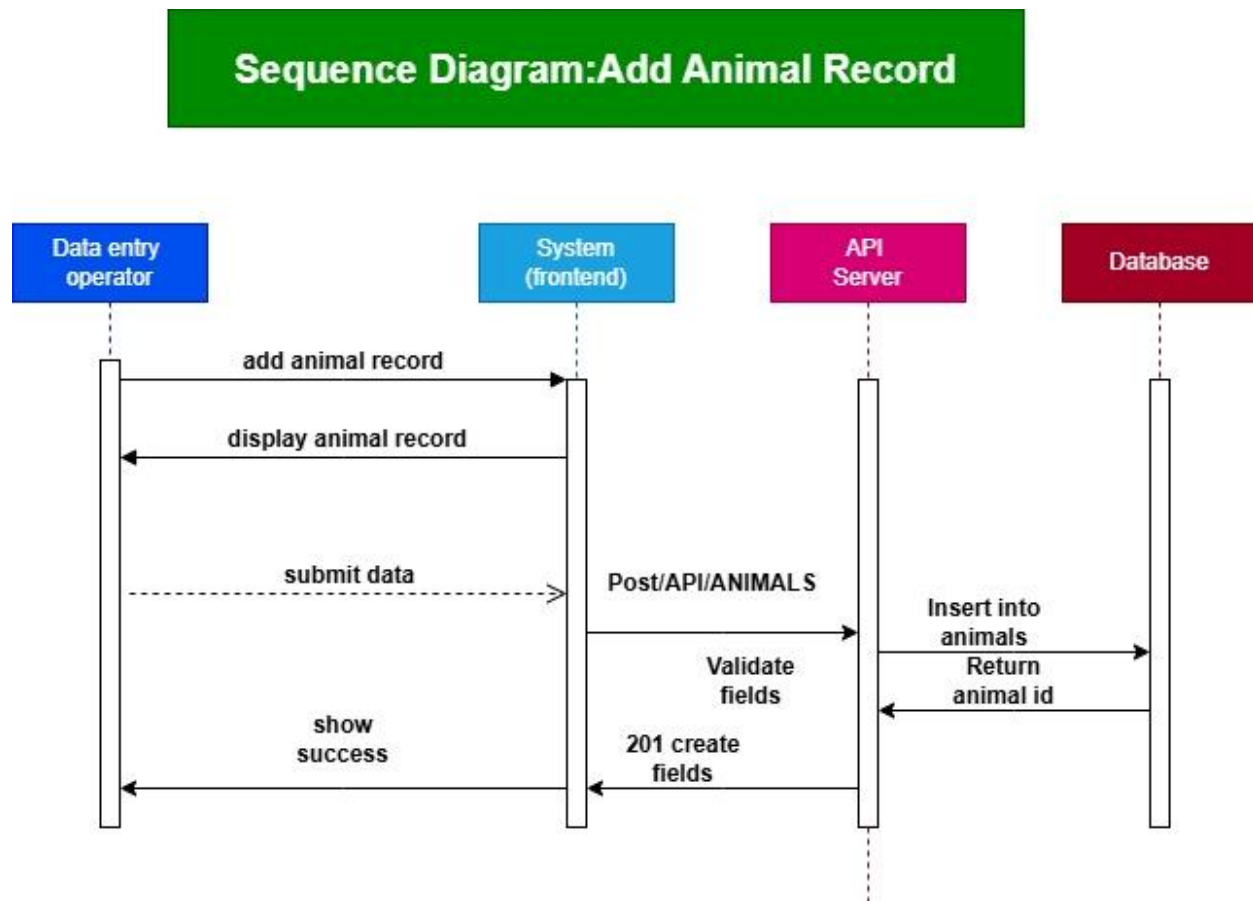
**Process 4 - User Processing:** An admin user can create, update, or delete user account information.

**Data Storage:** DS1 - Animals' table, DS2 - Movement's table, DS3 - DNA's table, DS4 - Health's table, DS5 - Users' table.

### 3.3 System Sequence Diagrams

System sequence diagram describes the interaction between actors and the system for key use cases:

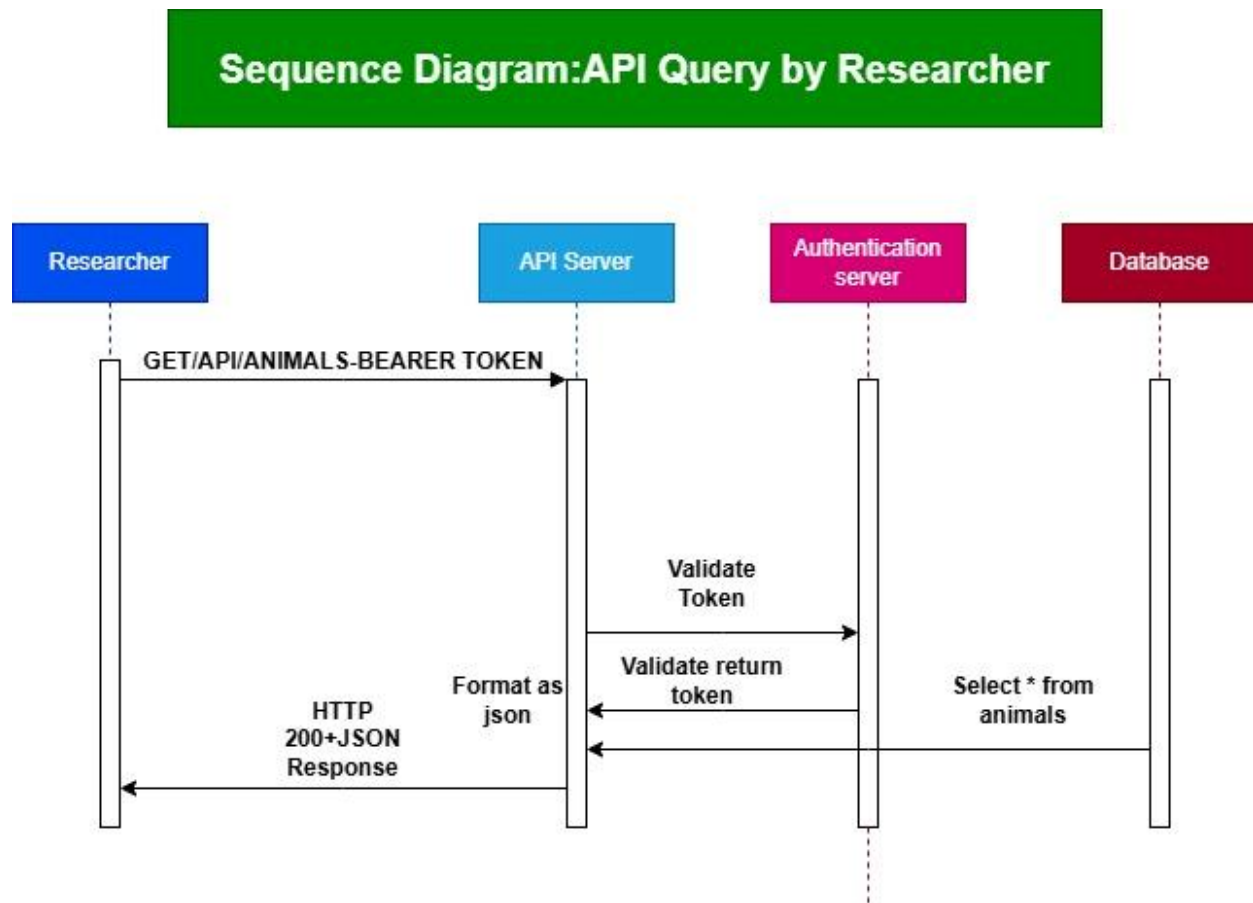
#### Sequence 1: Add Animal Record



**Figure 5: Sequence Diagram — Add Animal Record**

- Add Animal Form is done by Data Entry Operator.
- System displays the form with compulsory fields.
- Details are entered and managed by operator and submit button is pressed.
- System validates all mandatory and compulsory fields.
- System saves the record in Animals database table.
- System responds with success message containing the atomic or unique animal ID.

## Sequence 2: API Query by Researcher



**Figure 6: Sequence Diagram — API Query by Researcher**

HTTP GET request containing query parameters along with the Bearer Token is sent by the researcher.

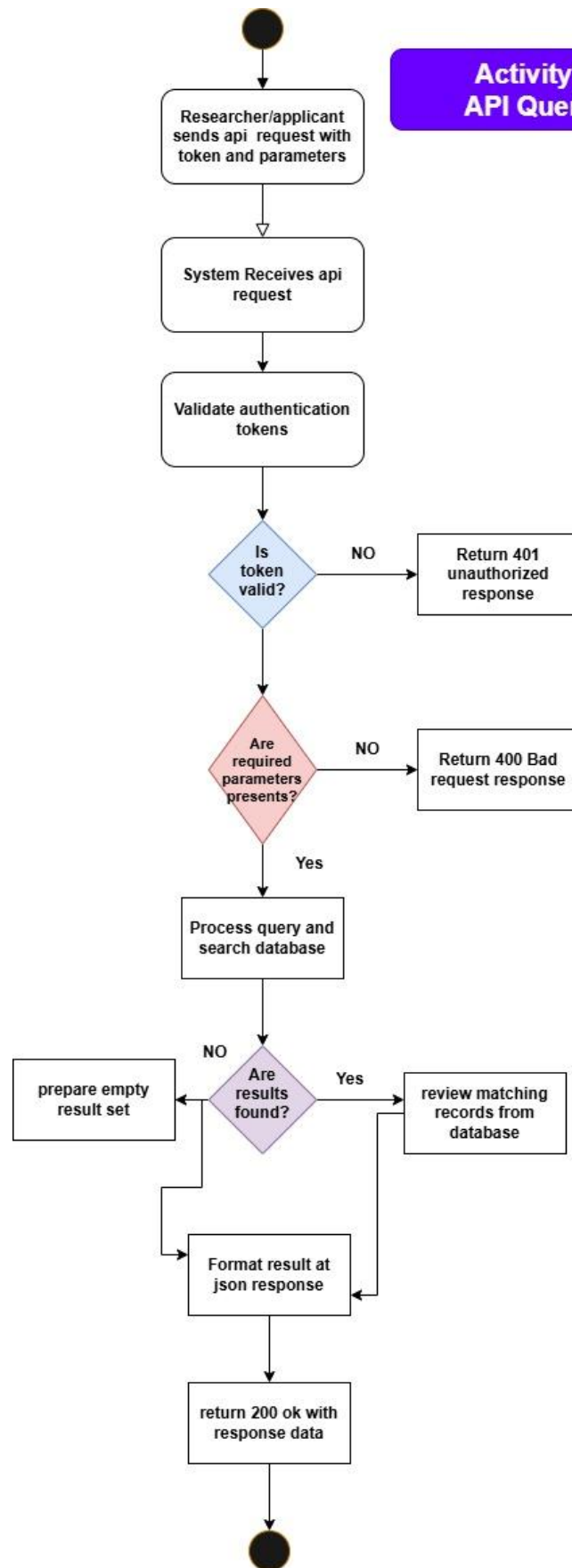
Authentication service authenticates the token.

Query is processed by the system on the basis of database.

Response from the system is in the form of JSON data and HTTP status code 200.

### 3.4 Activity Diagram

The activity diagram below describes the flow when a researcher queries data through the API:



**Figure 7: Activity Diagram — API Query Flow**

Begin: Researcher sends API request along with token.

Decision: Is the token valid? If No → Send 401 response.

If Yes → Proceed further.

Decision: Are all necessary parameters included in the query? If No → Send 400 response.

If Yes → Continue.

The system searches the database for the data.

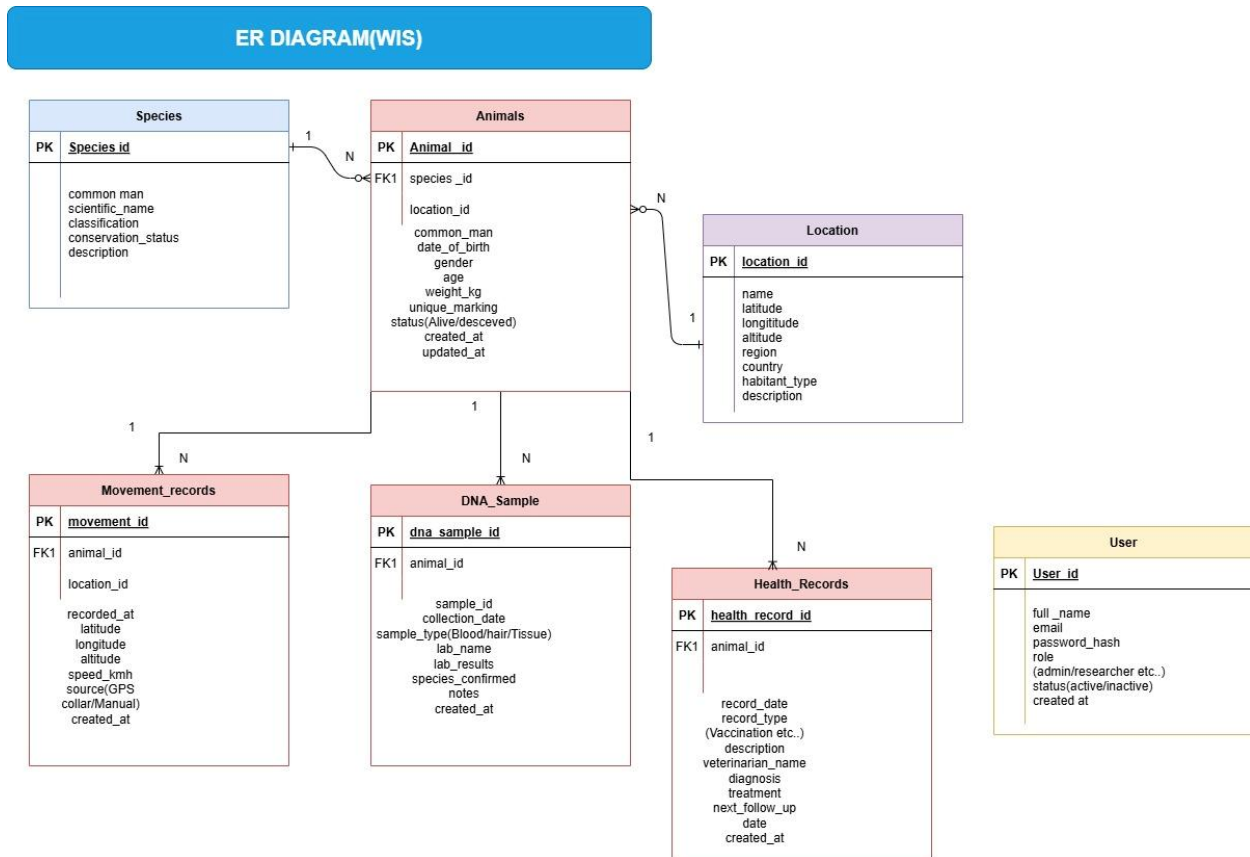
If not found → Return an empty result set array. If found → Format the response in JSON.

System sends 200 response.

End.

### 3.5 Database ER Diagram

The Entity-Relationship diagram tells us about the complete database schema for the Wildlife Information System, including all tables, primary keys (PK), foreign keys (FK), and their relationships.

**Figure 8: ER Diagram — Database Schema**

Key relationships: Animal references Species (FK). Movement\_record, DNA\_Sample, and Health\_Record all reference Animal (FK). Each User manages multiple operations tracked in audit logs.